

LAB 1 – USE-CASE FLOW SPECIFICATION

Problem Statement #12 – Healthcare & Telemedicine

Blood Bank Inventory & Emergency Donor Matcher

Use Case: UC-01 – Submit Emergency Blood Request

Primary Actor: Emergency Requester

Supporting Actors: Blood Bank Manager, Donor

Preconditions

1. The Emergency Requester has access to the Blood Bank system.
2. The system is operational.
3. Blood bank inventory information is available.
4. The requester has the required blood group, number of units, and emergency location details.

Postconditions

1. The emergency blood request is recorded in the system.
2. The available compatible blood inventory is checked.
3. If sufficient blood is unavailable, compatible eligible donors are searched.
4. Matching donors are notified if available.
5. The emergency request status is updated.

Main Success Scenario

1. The Emergency Requester selects the "**Submit Emergency Blood Request**" option from the system.
2. The system displays a form to enter the blood group, required number of units, emergency location, and urgency.
3. The Emergency Requester enters a valid blood group, required number of units, and emergency location.
4. The system validates the entered request details.
5. The system checks the blood bank inventory for compatible blood units.
6. Sufficient compatible blood units are available in the inventory.

7. The Blood Bank Manager allocates the required blood units to the emergency request.
 8. The system updates the blood inventory and records the allocation.
 9. The system displays the successful allocation status to the Emergency Requester.
 10. The emergency blood request is marked as "**Fulfilled.**"
 11. Use case ends successfully.
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Alternate Flow

6a. Insufficient Blood Available

6a1. The system determines that sufficient compatible blood units are not available in the inventory.

6a2. The system searches for eligible donors with a compatible blood group within a **10 km radius** of the emergency location.

6a3. The system checks the eligibility of the identified donors.

6a4. If eligible donors are found, the system sends emergency SMS notifications to the matching donors.

6a5. The system records the notification status and updates the emergency request as "**Awaiting Donor Response.**"

6a6. If no eligible donors are found, the system alerts the Blood Bank Manager about the shortage.

6a7. The system marks the request as "**Blood Shortage – Further Action Required.**"

6a8. Use case ends.